

Analysis of Education Data based on NSS 75th Round for Kerala and India

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Abstract

The National Sample Survey conducted its 75th Round of survey from 1st July 2017 to 30th June 2018 on ‘Social Consumption: Education’. The survey data on education is analysed here to examine the educational discontinuity and non-enrollment across all sectors and both genders among individuals from ages 3 to 35. The proportion of discontinuity and non-enrollment is estimated for Kerala and for all of India. Variation across various reasons of discontinuity is also discussed. The **usual monthly per capita expenditure (UMPCE)** quintile classes are obtained, and trends of the proportion across these classes are also observed.

1 Introduction

The National Sample Survey Office (NSSO), Ministry of Statistics and Programme Implementation (MOSPI), Government of India, since its inception in 1950 has been conducting nationwide integrated large scale sample surveys, employing scientific sampling methods, to generate data and statistical indicators on diverse socio-economic aspects. The NSS surveys are conducted by interviewing sample households selected through a scientific design and cover practically the entire geographical area of Indian Territory. In its 75th Round of survey, conducted during the period 1st July 2017 to 30th June 2018, NSSO carried out a survey on ‘Social Consumption: Education’. The last survey on this subject was conducted during 71st Round of NSS (January 2014 to June 2014).

1.1 Objectives of the Survey

The main objective of the survey on “Household Social Consumption: Education” conducted by NSSO is to build indicators on participation of the persons in the education system, expenditure incurred on education of the household members and on various aspects of those currently not attending education (i.e., for the persons who never enrolled or who ever enrolled but currently not attending education). The surveys on ‘Household Social Consumption: Education’ conducted by NSSO covers both qualitative and quantitative aspects related to educational attainment of the household members and educational services used by them. Qualitative aspects include literacy, educational level attained, type of institution, nature of institution, current attendance/enrolment, whether received free education, reason for never enrolled/ever enrolled but currently not attending, etc. On quantitative aspects, information is collected on expenditure incurred/to be incurred on education of the household member by the household itself, by other households or by any institutions/organizations other than Government.

1.2 Objectives of the Study

The following questions were studied in this report:

1. To obtain the state-wise, sector-wise, gender-wise estimates of the percentage of persons (age 3-35 years)
 - (a) ever enrolled but currently not attending any educational institution
 - (b) never-enrolled
 - (c) dropped out

for each State/UT in India and for Kerala in particular. Also, to obtain the estimates of the standard errors and the relative standard error in percentage.
2. To obtain the state-wise, sector-wise, gender-wise estimates of the Percentage of persons (age 3-35 years)
 - (a) ever enrolled but currently not attending any educational institution
 - (b) never-enrolled
 - (c) dropped out for each of the various reasons

considered in the survey. Also, to obtain the relative standard errors (RSE) in percentage.
3. To obtain the UMPCE (usual monthly per-capita expenditure) quintile classes for all of India and for Kerala for Rural and Urban categories of residence separately.
4. To obtain the estimates of the Percentage of persons (age 3-35 years)
 - (a) ever enrolled but currently not attending any educational institution
 - (b) never-enrolled
 - (c) dropped out

for each quintile class of UMPCE for all-India and for Kerala for rural/urban and male/female separately. Also, to obtain the relative standard errors (RSE) in percentage.
5. To obtain the All-India and state-wise estimates for Kerala for of the percentage of persons dropped out among ever enrolled persons of age 3 to 35 years for different levels of last enrolment for rural/urban categories of residence and for male/female separately. Also, to obtain the relative standard errors (RSE) in percentage.
6. To compare the state-wise estimates for Kerala with the national-level estimates.

2 Sampling Design

2.1 Geographic Coverage

The survey covers the whole of the Indian Union except except the villages in Andaman and Nicobar Islands which are difficult to access.

2.2 Universe

The survey used the interview method of data collection from a sample of randomly selected households and members of the household.

2.3 Period of Survey and Work Programme

The period of survey is of twelve months duration starting on 1st July 2017 and ending on 30th June 2018. The survey period of the round is divided into four sub-rounds of three months' duration each as follows:

sub-round 1 : July - September 2017
sub-round 2 : October - December 2017
sub-round 3 : January - March 2018
sub-round 4 : April - June

In each of these four sub-rounds equal number of sample villages/ blocks (FSUs) is allotted for survey with a view to ensure uniform spread of sample FSUs over the entire survey period. Because of the arduous field conditions, this restriction is not strictly enforced in Andaman and Nicobar Islands, Lakshadweep, Ladakh region (Leh and Kargil districts) of Jammu & Kashmir and rural areas of Arunachal Pradesh and Nagaland.

2.4 Participation of States

All the States and Union Territories except Andaman & Nicobar Islands, Chandigarh, Dadra & Nagar Haveli and Lakshadweep participated. Following is the matching pattern of the participating States/ UTs.

Table 1: Extent of Matching by State/UT

State/UT	Extent of Matching
Nagaland (U)	Triple
Andhra Pradesh, Manipur, Telangana, Chhattisgarh	Double
Gujarat, Maharashtra (U)	One and a Half
Remaining States/UTs	Equal

Thus the matching pattern of Kerala is 1.

2.5 Sampling Design

2.5.1 Outline of sample design

A stratified multi-stage design has been adopted for the 75th round survey. The first stage units (FSU) are the Census villages (Panchayat wards for Kerala) in the rural sector and Urban Frame Survey (UFS) blocks in the urban sector. The ultimate stage units (USU) are households in both the sectors. In the case of large FSUs, one intermediate stage of sampling is the selection of two hamlet-groups (hgs)/ sub-blocks (sbs) from each rural/ urban FSU.

2.5.2 Sampling Frame for First Stage Units

For the rural sector, the list of 2011 Population Census villages constitutes the sampling frame. However, for Kerala, the latest available updated list of Panchayat wards constitutes the sampling frame. For the urban sector, the latest available list of UFS blocks has been considered as the sampling frame.

2.5.3 Stratification

1. Each district is a stratum. Within each district of a State/UT, generally speaking, two basic strata have been formed:

- (a) rural stratum comprising of all rural areas of the district
- (b) urban stratum comprising of all the urban areas of the district

However, within the urban areas of a district, if there are one or more towns with population one million or more as per Census 2011, each of them formed as a separate basic stratum and the remaining urban areas of the district have been considered as another basic stratum.

2. In the case of rural sectors of Nagaland another special stratum has been formed within the State consisting of all the villages which are difficult to access.

2.5.4 Sub-stratification

1. **Rural sector:** If 'r' be the sample size allocated for a rural stratum, 'r/4' sub-strata are formed in that stratum. The villages within a stratum (district) as per frame are first arranged in ascending order of population. Then sub-strata 1 to 'r/4' have been demarcated in such a way that each sub-stratum comprises a group of villages of the arranged frame and has more or less equal population.
2. **Urban sector:** Each urban stratum has been divided into two parts - 'Affluent part' consisting of UFS blocks identified as 'Affluent Area' and 'Non-Affluent part' consisting of the remaining UFS blocks within the stratum. Sample allocation of a stratum is then allocated to Affluent and Non-Affluent parts in proportion to total number of households in the UFS blocks with double weightage to Affluent part subject to a maximum of 8 FSUs in 'Affluent part' of any stratum. If 'u' be the sample size allocated for an urban stratum consisting of both affluent area UFS Blocks and non affluent area UFS Blocks. 'u/4' sub-strata are formed within each stratum. Out of these 'u/4' substrata, the first two sub-strata '01' and '02' are earmarked for those UFS blocks which are identified as 'Affluent Area' and the remaining sub-strata, '03', '04',.... and so on, are assigned to the nonaffluent UFS blocks. If any stratum does not have any Affluent Area UFS block then also the substratum number starts from '03' for that stratum. For all strata, if $u/4 > 1$, implying formation of 2 or more sub-strata, all the UFS blocks within the stratum are first arranged in ascending order of total number of households in the UFS Blocks as per the latest UFS phase. Then sub-strata 1 to 'u/4' are demarcated in such a way that each sub-stratum has more or less equal number of households. This procedure has been done separately for Affluent-part and Non-Affluent part of the stratum (if two sub-strata are required to be formed in the Affluent part). The following three cases arise while doing the sub-stratification:
 - (a) If there is no 'Affluent Area' UFS block in the stratum, all the UFS blocks are divided into 'u/4' sub-strata and numbered as '03', '04', ..., '(u/4)+2'.
 - (b) If only one sub-stratum is formed with the 'Affluent Area' UFS blocks, then all the remaining non-affluent blocks are divided into '(u-4)/4' sub-strata and numbered as '03', '04',....., '(u/4)+1'.
 - (c) If two sub-strata are formed with the 'Affluent Area' blocks, then all the remaining non-affluent UFS blocks are divided into '(u-8)/4' sub-strata and numbered as '03', '04',....., 'u/4'.

2.5.5 Total sample size (FSUs)

14300 FSUs have been allocated for the central sample at all-India level. For the state sample, there are 16492 FSUs allocated for all-India.

2.5.6 Allocation of total sample to States and UTs

The total number of sample FSUs has been allocated to the States and UTs in proportion to population as per Census 2011 subject to a minimum sample allocation to each State/UT.

2.5.7 Allocation of State/ UT level sample to rural and urban sectors

State/ UT level sample size has been allocated between two sectors in proportion to population as per Census 2011 with 1.5 weightage to urban sector depending on population share. A minimum of 16 FSUs (minimum 8 each for rural and urban sector separately) have been allocated to each State/ UT.

2.5.8 Allocation to strata

Within each sector of a State/ UT, the respective sample size has been allocated to the different strata in proportion to the population as per Census 2011. Stratum level allocation has been adjusted to multiples of 4 with a minimum sample size of 4. For special stratum formed in rural areas of Nagaland as discussed in subsection 2.5.3 (2), 12 FSUs have been allocated.

2.5.9 Allocation to sub-strata

Allocation for each sub-stratum has been made as 4 in both rural and urban sectors.

2.5.10 Selection of FSUs

1. For the rural sector, from each stratum/sub-stratum, required number of sample villages has been selected by Probability Proportional to Size With Replacement (PPSWR), size being the population of the village as per Census 2011.
2. For the urban sector, from each stratum/sub-stratum, FSUs have been selected by Probability Proportional to Size With Replacement (PPSWR), size being the number of households of the UFS Block. Both rural and urban samples are drawn in the form of two independent sub-samples.
3. Selection of hamlet-groups/ sub-blocks - important steps
4. Criterion for hamlet-group/ sub-block formation: After identification of the boundaries of the FSU, it is to be determined whether listing is to be done in the whole sample FSU or not. In case the approximate present population of the selected FSU is found to be 1200 or more, it is divided into a suitable number (say, D) of 'hamlet-groups' in the rural sector and 'sub-blocks' in the urban sector by more or less equalising the population as stated below. approximate present population of the sample FSU no. of hgs/sbs

Table 2: Approximate Present Population of the Sample Village and Number of Hamlet-Groups/Sub-Blocks Formed

Approximate Present Population of the Sample Village	Number of HGs/Sub-Blocks Formed
Less than 1200	1
1200 to 1799	3
1800 to 2399	4
2400 to 2999	5
3000 to 3599	6
<i>and so on</i>	

For rural areas of Himachal Pradesh, Sikkim, Andaman & Nicobar Islands, Uttarakhand (except four districts Dehradun, Nainital, Hardwar and Udham Singh Nagar), Punch, Rajouri, Udhampur, Reasi, Doda, Kishtwar, Ramban, Ladakh region (Leh and Kargil districts) of Jammu and Kashmir and Idukki district of Kerala, the number of hamlet-groups to be formed as follows:

Table 3: Approximate Present Population of the Sample Village and Number of Hamlet-Groups (HGs) Formed

Approximate Present Population of the Sample Village	Number of HGs Formed
Less than 600	1
600 to 899	3
900 to 1199	4
1200 to 1499	5
1500 to 1799	6
<i>and so on</i>	

2.5.11 Formation and selection of hamlet-groups/ sub-blocks

In case hamlet-groups/ sub-blocks are to be formed in the sample FSU, the same should be done by more or less equalising population. Note that while doing so, it is to be ensured that the hamlet-groups/ sub-blocks formed are clearly identifiable in terms of physical landmarks. Two hamlet-groups (hg)/ sub-blocks (sb) are to be selected from a large FSU wherever hamletgroups/ sub-blocks have been formed in the following manner - one hg/ sb with maximum percentage share of population will always be selected and termed as hg/ sb 1; one more hg/ sb will be selected from the remaining hgs/ sbs by simple random sampling (SRS) and termed as hg/ sb 2. Listing and selection of the households are being done independently in the two selected hamlet-groups/ subblocks. The FSUs without hg/ sb formation will be treated as sample hg/ sb number 1.

2.5.12 Listing of households

Having determined the hamlet-groups/ sub-blocks, i.e. area(s) to be considered for listing, the next step is to list all the households. The hamlet-group/ sub-block with hg/sb number 1 will be considered for listing first, to be followed by the listing of households of the hg/sb number 2.

2.5.13 Formation of second stage strata (SSS) and allocation of households in different SSS

Table 4: Formation of second stage strata and allocation of households

SSS	Composition of SSS	number of households to be surveyed	
		FSU without hg/sb formation	FSU with hg/sb per hg/sb)
SSS 1	Households with at least one student* having disability (possessing a certificate of disability)	2	1
SSS 2	From the remaining, households with at least one student* receiving technical/professional education	2	1
SSS 3	from the remaining, households having at least one student* receiving general education	2	1
SSS 4	other households	2	1

**Persons aged 3-35 years and currently attending education are considered as students.*

2.5.14 Selection of households

From each SSS, the sample households for each of the schedules are selected by SRSWOR.

3 Estimation Procedure

3.1 Notations

s	= subscript for s -th stratum
t	= subscript for t -th sub-stratum
m	= subscript for sub-sample ($m = 1, 2$)
i	= subscript for i -th FSU [village (panchayat ward)/ block]
d	= subscript for a hamlet-group/ sub-block ($d = 1, 2$)
j	= subscript for j -th second stage stratum in an FSU/hg/sb
k	= subscript for k -th sample household within an FSU/ hg/sb
D	= total number of hg's/ sb's formed in the sample FSU
D^*	= 0 if $D = 1$ = $(D - 1)$ for FSUs with $D > 1$
Z	= total size of a sub-stratum (sum of sizes for all the FSUs of a sub-stratum)
z	= size of sample FSU used for selection
n	= number of sample FSUs surveyed including 'uninhabited' and 'zero cases' but excluding casualty for a particular sub-sample and sub-stratum
H	= total number of households listed in a second-stage stratum of an FSU / hamlet-group or sub-block of sample FSU
h	= number of households surveyed in a second-stage stratum of an FSU / hamlet-group or sub-block of sample FSU
x, y	= observed value of characteristics x, y under estimation
$\hat{X}, \hat{Y}$	= estimate of population total X, Y for the characteristics x, y

Under the above symbols, $y_{stmidjk}$ = observed value of the characteristic y for the k -th household of the j -th second stage stratum of the d -th hg/sb of the i -th FSU belonging to the m -th sub-sample for the t -th sub-stratum of s -th stratum.

However, for ease of understanding, a few symbols have been suppressed in following paragraphs where they are obvious.

3.2 Formulae for Estimation of Aggregates for a particular sub-sample and stratum $\times$ substratum

Rural/ Urban:

(i) For j -th second-stage stratum of a stratum $\times$ sub-stratum:

$$\hat{Y}_j = \frac{Z}{n_j} \sum_{i=1}^{n_j} \frac{1}{z_i} \left[\frac{H_{i1j}}{h_{i1j}} \sum_{k=1}^{h_{i1j}} y_{i1jk} + D_i^* \times \frac{H_{i2j}}{h_{i2j}} \sum_{k=1}^{h_{i2j}} y_{i2jk} \right] \quad (1)$$

(ii) For all second-stage strata combined:

$$\hat{Y} = \sum_j \hat{Y}_j \quad (2)$$

3.3 Overall Estimate for Aggregates for a sub-stratum

Overall estimate for aggregates for a sub-stratum ($\hat{Y}_{st}$) based on two sub-samples in a sub-stratum is obtained as:

$$\hat{Y}_{st} = \frac{1}{2} \sum_{m=1}^2 \hat{Y}_{stm}$$

3.4 Overall Estimate for Aggregates for a stratum

Overall estimate for a stratum ($\hat{Y}_s$) will be obtained as

$$\hat{Y}_s = \sum_t \hat{Y}_{st}$$

3.5 Overall Estimate of Aggregates at State/UT/all-India level

The overall estimate $\hat{Y}$ at the State/ UT/ all-India level is obtained by summing the stratum estimates $\hat{Y}_s$ over all strata belonging to the State/ UT/ all-India.

3.6 Estimates of Ratios

Let $\hat{Y}$ and $\hat{X}$ be the overall estimates of the aggregates Y and X for two characteristics y and x respectively at the State/ UT/ all-India level.

Then the combined ratio estimate ($\hat{R}$) of the ratio ($R = \frac{Y}{X}$) will be obtained as $\hat{R} = \frac{\hat{Y}}{\hat{X}}$.

3.7 Estimates of Error

The estimated variances of the above estimates will be as follows:

3.7.1 For aggregate $\hat{Y}$

$$\hat{V}ar(\hat{Y}) = \sum_s \hat{V}ar(\hat{Y}_s) = \sum_s \sum_t \hat{V}ar(\hat{Y}_{st}) \text{ where } \hat{V}ar(\hat{Y}_{st}) \text{ is given by}$$

$$\hat{V}ar(\hat{Y}_{st}) = \frac{1}{4} \left[\hat{Y}_{st1} - \hat{Y}_{st2} \right]^2, \text{ where } \hat{Y}_{st1} \text{ and } \hat{Y}_{st2} \text{ are the estimates for sub-sample 1 and sub-sample 2 respectively for stratum 's' and sub-stratum 't'.$$

3.7.2 For ratio $\hat{R}$

$$MSE(\hat{R}) = \frac{1}{4\hat{X}^2} \sum_s \sum_t \left[\left(\hat{Y}_{st1} - \hat{Y}_{st2} \right)^2 + \hat{R}^2 \left(\hat{X}_{st1} - \hat{X}_{st2} \right)^2 - 2\hat{R} \left(\hat{Y}_{st1} - \hat{Y}_{st2} \right) \left(\hat{X}_{st1} - \hat{X}_{st2} \right) \right]$$

3.8 Estimates of Relative Standard Error (RSE)

$$R\hat{S}E(\hat{Y}) = \frac{\sqrt{\hat{V}ar(\hat{Y})}}{\hat{Y}} \times 100$$

$$R\hat{S}E(\hat{R}) = \frac{\sqrt{MSE(\hat{R})}}{\hat{R}} \times 100$$

3.9 Multipliers

Instead of individually estimating at each level and then using estimates from each level to compute the estimate at the next level, we can combine all the formulae into a weighted sum of the observations. These weights are called multipliers, which are determined by the estimation formulae depending on the type of sampling adopted at each stage. The formulae for multipliers at stratum/sub-stratum/second-stage stratum level for a sub-sample and schedule type are given below:

Table 5: Formulae for multipliers

sector	formula for multipliers (hg / sb 1)	formula for multipliers (hg / sb 2)
rural/ urban	$\frac{Z_{st}}{n_{stmj}} \times \frac{1}{z_{stmi}} \times \frac{H_{stmilj}}{h_{stmilj}}$	$\frac{Z_{st}}{n_{stmj}} \times \frac{1}{z_{stmi}} \times D *_{stmi} \times \frac{H_{stmil2j}}{h_{stmil2j}}$

Explanation for terms in multiplier:

1. Note that in **PPSWR**, the estimate for population total is $\sum_{r=1}^n \frac{y_r}{np_r}$. Since FSUs are selected by PPSWR, the term $\frac{Z_{st}}{n_{stmj} \times z_{stmj}}$ appears, since n_{stmj} is like sample size, and $\frac{z_{stmj}}{Z_{st}}$ is the proportion.
2. Note that in **SRSWOR**, the estimate for population total is $\sum_{r=1}^n \frac{Ny_r}{n}$. Since households are selected via SRSWOR, the term $\frac{H_{stmilj}}{h_{stmilj}}$ appears, as H total number of households listed and h represents number of households surveyed (sample size).
3. Similarly, the multiplier for the cases when there is more than 1 hg/sb is calculated based on these ideas.

4 Results

4.1 Drop outs

According to the report on **Key Indicators of Household Social Consumption on Education in India** of the NSSO 75th Round, an ever-enrolled person was considered as dropped out if he/ she did not complete the last level of education for which he/she enrolled and currently not attending any educational institution for reasons other than ‘completion of the desired level of education’.

4.2 Estimates and Errors

Table 6: Dropout Rates Among 3-35 Years of Age for Kerala

Sector	Sex	Estimate (%)	Standard Error	RSE
Rural	Male	0.002	0.0004	25.00
Rural	Female	0.044	0.0081	18.35
Urban	Male	0.105	0.0012	1.19
Urban	Female	0.131	0.0050	3.83

Table 7: Dropout Rates Among 3-35 Years of Age for all India

Sector	Sex	Estimate (%)	Standard Error	RSE
Rural	Male	0.10	0.004	3.97
Rural	Female	0.14	0.005	3.74
Urban	Male	0.09	0.003	3.31
Urban	Female	0.09	0.005	5.33

Table 8: Ever Enrolled but Currently Not Attending Education for Age 3-35 for Kerala

Sector	Sex	Estimate (%)	Standard Error(%)	RSE
Rural	Male	38.65	0.65	1.69
Rural	Female	46.06	0.73	1.59
Urban	Male	41.85	0.81	1.93
Urban	Female	46.22	0.86	1.86

Table 9: Ever Enrolled but Currently Not Attending Education for Age 3-35 for all India

Sector	Sex	Estimate (%)	Standard Error(%)	RSE
Rural	Male	41.32	0.41	1.00
Rural	Female	40.04	0.39	0.98
Urban	Male	46.20	0.46	1.00
Urban	Female	47.77	0.48	1.00

Table 10: Never Enrolled for Kerala

Sector	Sex	Estimate (%)	Standard Error	RSE
Rural	Male	3.86	0.13	3.37
Rural	Female	4.35	0.14	3.23
Urban	Male	3.88	0.15	3.83
Urban	Female	2.24	0.11	5.02

Table 11: Never Enrolled for all India

Sector	Sex	Estimate (%)	Standard Error	RSE
Rural	Male	12.62	0.13	1.01
Rural	Female	19.27	0.19	1.01
Urban	Male	7.09	0.09	1.25
Urban	Female	9.62	0.12	1.27

Table 12: Reasons for Non-Enrollment by Sector for Males (all-India)

Sector	Reason	Estimate (%)	SE	RSE
Rural	Not interested in education	26.22	1.92	7.31
Rural	Financial constraints	5.84	0.60	10.34
Rural	Engaged in domestic activities	0.79	0.03	4.12
Rural	Engaged in economic activities	1.38	0.16	11.29
Rural	Timings not suitable	0.16	0.04	25.17
Rural	No tradition in community	1.42	0.06	4.12
Rural	Others	64.20	2.55	3.97
Urban	Not interested in education	17.56	1.51	8.61
Urban	Financial constraints	10.38	0.97	9.34
Urban	Timings not suitable	0.89	0.23	25.35
Urban	No tradition in community	0.22	0.06	25.46
Urban	Others	70.95	3.30	4.65

Table 13: Reasons for Discontinuance by Sector for Males(all-India)

Sector	Reason	Estimate (%)	SE	RSE
Rural	Not interested in education	22.14	0.71	3.20
Rural	Financial constraints	19.86	0.63	3.18
Rural	Engaged in domestic activities	8.19	0.37	4.55
Rural	Engaged in economic activities	29.57	0.90	3.04
Rural	School is far off	0.51	0.09	17.78
Rural	Unable to cope up/failure in studies	3.48	0.19	5.58
Rural	Completed desired level/class	8.13	0.48	5.85
Rural	Preparation for competitive exam	2.52	0.23	9.18
Rural	Others	5.59	0.40	7.20
Urban	Not interested in education	26.09	1.03	3.96
Urban	Financial constraints	16.03	0.75	4.65
Urban	Engaged in domestic activities	7.73	0.32	4.18
Urban	Engaged in economic activities	35.26	1.16	3.28
Urban	School is far off	0.02	0.001	2.85
Urban	Language/medium unfamiliar	0.70	0.16	22.69
Urban	Unable to cope up/failure in studies	1.55	0.05	3.42
Urban	Unfriendly atmosphere at school	0.61	0.15	25.03
Urban	Completed desired level/class	6.59	0.38	5.73
Urban	Preparation for competitive exam	3.23	0.09	2.92
Urban	Others	2.19	0.07	2.97

Table 14: Reasons for Dropping out by Sector for Males (All-India)

Sector	Reason	Estimate (%)	SE	RSE
Rural	Not interested in education	29.21	1.76	6.03
Rural	Financial constraints	16.28	1.68	10.34
Rural	Engaged in domestic activities	11.68	1.30	11.11
Rural	Engaged in economic activities	26.00	1.89	7.26
Rural	Unable to cope up/failure in studies	14.78	0.97	6.54
Rural	Others	2.04	0.11	5.43
Urban	Not interested in education	23.18	1.96	8.44
Urban	Financial constraints	19.16	1.99	10.41
Urban	Engaged in domestic activities	4.68	0.51	10.97
Urban	Engaged in economic activities	38.93	2.39	6.15
Urban	Language/medium unfamiliar	3.09	0.77	24.91
Urban	Unable to cope up/failure in studies	7.29	0.45	6.17
Urban	Preparation for competitive exam	0.23	0.01	5.86
Urban	Others	3.44	0.20	5.86

Table 15: Reasons for Never Enrollment by Sectors for Females (All-India)

Sector	Reason	Estimate (%)	SE	RSE
Rural	Not interested in education	19.32	0.85	4.41
Rural	Financial constraints	5.94	0.55	9.21
Rural	Engaged in domestic activities	5.75	0.41	7.21
Rural	Engaged in economic activities	2.92	0.39	13.46
Rural	School is far off	0.43	0.10	22.96
Rural	Language/medium unfamiliar	0.01	0.0003	2.75
Rural	No tradition in community	3.26	0.36	11.13
Rural	Others	62.37	1.57	2.52
Urban	Not interested in education	14.95	1.02	6.83
Urban	Financial constraints	15.35	0.87	5.65
Urban	Engaged in domestic activities	4.19	0.61	14.45
Urban	Engaged in economic activities	1.23	0.21	17.43
Urban	Timings not suitable	1.04	0.12	11.07
Urban	Route not safe	0.79	0.20	25.19
Urban	No tradition in community	4.19	0.71	16.90
Urban	Others	58.24	2.36	4.04
Urban	Marriage	0.02	0.001	4.42

Table 16: Reasons for Discontinuance by Sectors for Females (All-India)

Sector	Reason	Estimate (%)	SE	RSE
Rural	Not interested in education	14.03	0.44	3.17
Rural	Financial constraints	15.52	0.53	3.44
Rural	Engaged in domestic activities	32.62	0.92	2.82
Rural	Engaged in economic activities	1.68	0.17	10.15
Rural	School is far off	6.03	0.60	9.96
Rural	Language/medium unfamiliar	0.32	0.08	24.88
Rural	Route not safe	0.01	0.002	25.11
Rural	Unable to cope up/failure in studies	2.44	0.12	5.07
Rural	Unfriendly atmosphere at school	0.92	0.02	2.65
Rural	Completed desired level/class	7.36	0.39	5.37
Rural	Preparation for competitive exam	0.14	0.03	17.85
Rural	Non-availability of girls' toilet	0.33	0.009	2.65
Rural	Others	8.28	0.94	11.36
Rural	Marriage	10.31	0.54	5.21
Urban	Not interested in education	22.34	0.81	3.65
Urban	Financial constraints	8.30	0.35	4.26
Urban	Engaged in domestic activities	34.17	1.06	3.11
Urban	Engaged in economic activities	6.24	0.43	6.93
Urban	School is far off	2.67	0.29	10.68
Urban	Language/medium unfamiliar	0.06	0.002	2.71
Urban	Route not safe	0.05	0.001	2.71
Urban	Unable to cope up/failure in studies	1.96	0.12	6.36
Urban	Unfriendly atmosphere at school	0.58	0.15	25.24
Urban	Completed desired level/class	6.35	0.37	5.76
Urban	Preparation for competitive exam	1.80	0.17	9.29
Urban	Others	3.00	0.23	7.77
Urban	Marriage	12.48	0.44	3.50

Table 17: Reasons for Dropping out by Sectors for Females (All-India)

Sector	Reason	Estimate (%)	SE	RSE
Rural	Not interested in education	16.21	1.06	6.54
Rural	Financial constraints	12.08	1.24	10.22
Rural	Engaged in domestic activities	39.38	1.69	4.30
Rural	Engaged in economic activities	1.33	0.21	15.94
Rural	School is far off	8.68	2.02	23.31
Rural	Route not safe	0.04	0.01	25.31
Rural	Unable to cope up/failure in studies	8.90	0.47	5.26
Rural	Others	4.85	0.80	16.46
Rural	Marriage	8.52	0.79	9.24
Urban	Not interested in education	30.50	2.95	9.66
Urban	Financial constraints	6.87	0.94	13.71
Urban	Engaged in domestic activities	28.15	1.78	6.34
Urban	Engaged in economic activities	1.83	0.34	18.75
Urban	School is far off	1.25	0.08	6.34
Urban	Unable to cope up/failure in studies	9.96	0.88	8.83
Urban	Others	4.29	0.27	6.23
Urban	Marriage	17.16	1.23	7.15

Table 18: Reasons for Discontinuation of Education by Sectors among Males in Kerala

Sector	Reason	Percentage (%)	SE	RSE (%)
Rural	Financial Constraints	1.21	0.37	30.45
Rural	Not Interested in Education	40.14	7.18	17.90
Rural	Engaged in Economic Activities	35.54	6.33	17.82
Rural	Completed Desired Level/Class	23.11	5.91	25.57
Urban	Financial Constraints	15.28	3.10	20.27
Urban	Not Interested in Education	3.51	1.09	31.15
Urban	Engaged in Economic Activities	1.76	0.55	31.15
Urban	Unable to Cope up with Studies/Failure in Studies	77.44	15.70	20.27
Urban	Others	2.01	0.41	20.27

Table 19: Reasons for Discontinuation of Education by Sectors among Females in Kerala

Sector	Reason	Percentage (%)	SE	RSE (%)
Rural	Preparation for Competitive Examination	38.49	7.31	18.99
Rural	Marriage	61.51	7.55	12.28
Urban	Not Interested in Education	2.91	0.90	30.94
Urban	Financial Constraints	18.30	4.44	24.25
Urban	Unable to Cope up with Studies/Failure in Studies	73.54	14.12	19.20
Urban	Completed Desired Level/Class	1.67	0.51	30.86
Urban	Marriage	3.58	0.79	22.01

In the NSS data, only five people from Kerala from our sample had never enrolled for education in the age group of 3-35 years. Thus, we are unable to guarantee stability of computed estimates for the non-enrollment in Kerala.

Due to the sparsity of data on dropping out in Kerala, we are unable to reliably compute the estimates and relative standard errors across sectors and gender.

Table 20: Percentage of persons dropped out among ever enrolled persons of age 3 to 35 years for different levels of last enrolment(all-India)

Level of last enrolment	Rural			Urban		
	Male	Female	RSE (%)	Male	Female	RSE (%)
Pre-primary	5.1	9.6	8.4	3.3	4.4	9.1
Primary	9.1	12.3	6.2	6.9	9.0	6.8
Upper primary / Middle	17.5	19.1	4.8	14.9	15.5	4.5
Secondary	21.4	20.2	3.9	17.6	16.6	3.6
Higher secondary	10.7	9.7	5.1	8.8	7.7	5.4
Diploma / certificate (below graduate)	1.7	2.1	12.8	9.7	16.4	7.2
Graduation	5.6	6.7	7.3	4.4	3.8	9.8
Post-graduation and above	3.9	2.8	10.6	1.6	1.5	11.4
Diploma / certificate (graduate and above)	10.1	4.0	9.2	4.2	1.9	7.2
All levels	13.2	14.7	2.9	9.5	9.7	3.1

Table 21: Percentage of persons dropped out among ever enrolled persons of age 3 to 35 years for different levels of last enrolment(Kerala)

Level of last enrolment	Rural			Urban		
	Male	Female	RSE (%)	Male	Female	RSE (%)
Pre-primary	4.6	8.2	7.9	3.1	3.9	8.8
Primary	8.7	11.5	6.0	6.3	8.1	6.4
Upper primary / Middle	16.2	18.0	4.6	13.8	14.5	4.4
Secondary	19.7	18.5	3.8	16.3	15.2	3.5
Higher secondary	9.6	8.9	4.9	8.0	7.1	5.2
Diploma / certificate (below graduate)	1.5	1.9	12.2	8.5	14.7	6.9
Graduation	5.1	6.1	7.0	4.0	3.6	9.3
Post-graduation and above	3.5	2.6	10.2	1.4	1.3	11.0
Diploma / certificate (graduate and above)	9.2	3.8	8.8	3.8	1.8	7.0
All levels	12.5	13.9	2.8	9.0	9.3	3.0

4.3 Quintile Classes

Table 22: Quintile Classes - All India

Quintile classes	Rural		Urban	
	Lower	Upper	Lower	Upper
1	0	1000.000	0	1800.000
2	1000.000	1333.333	1800.000	2500.000
3	1333.333	1648.571	2500.000	3333.333
4	1648.571	2100.000	3333.333	4750.000
5	2100.000	-	4750.000	-

Table 23: Quintile Classes - Kerala

Quintile classes	Rural		Urban	
	Lower	Upper	Lower	Upper
1	0	1900.000	0	2000.000
2	1900.000	2250.000	2000.000	2533.333
3	2250.000	2896.000	2533.333	3200.000
4	2896.000	3666.667	3200.000	4435.000
5	3666.667	-	4435.000	-

4.4 Estimates for Quintile Classes

Table 24: Estimates and Standard Errors for Never Enrollment by Quintile Classes-India Rural

Quintile Class	Male			Female		
	Estimate (%)	SE (%)	RSE	Estimate (%)	SE (%)	RSE
1	22.34	0.31	1.39	31.73	0.45	1.42
2	20.80	0.24	1.17	28.14	0.31	1.10
3	18.94	0.26	1.39	26.86	0.35	1.32
4	14.66	0.19	1.32	19.69	0.26	1.32
5	11.25	0.19	1.72	15.81	0.29	1.81

Table 25: Estimates and Standard Errors for Ever Enrolled but Currently not attending by Quintile Classes-India Rural

Quintile Class	Male			Female		
	Estimate (%)	SE (%)	RSE	Estimate (%)	SE (%)	RSE
1	45.29	0.55	1.22	40.32	0.49	1.21
2	46.93	0.52	1.11	42.79	0.46	1.08
3	48.99	0.52	1.05	44.47	0.48	1.08
4	50.59	0.57	1.14	48.66	0.55	1.13
5	53.75	0.91	1.69	53.54	0.90	1.68

Table 26: Estimates and Standard Errors for Dropping Out by Quintile Classes—India Rural

Quintile Class	Male			Female		
	Estimate (%)	SE (%)	RSE	Estimate (%)	SE (%)	RSE
1	45.11	0.55	1.22	40.20	0.49	1.21
2	46.64	0.52	1.11	42.59	0.46	1.08
3	48.51	0.51	1.06	44.22	0.48	1.08
4	49.89	0.57	1.14	48.20	0.54	1.13
5	52.39	0.89	1.70	52.63	0.89	1.69

Table 27: Estimates and Standard Errors for Never Enrollment by Quintile Classes—Kerala Rural

Quintile Class	Male			Female		
	Estimate (%)	SE (%)	RSE	Estimate (%)	SE (%)	RSE
1	12.30	1.18	9.56	7.02	0.54	7.63
2	7.31	1.17	15.97	15.58	1.65	10.61
3	8.72	0.73	8.40	6.48	0.56	8.65
4	6.79	0.48	7.10	12.81	0.62	4.81
5	4.33	0.13	2.97	4.62	0.19	4.20

Table 28: Estimates and Standard Errors for Ever Enrolled but Currently not attending by Quintile Classes-Kerala Rural

Quintile Class	Male			Female		
	Estimate (%)	SE (%)	RSE	Estimate (%)	SE (%)	RSE
1	57.39	5.27	9.17	53.22	3.94	7.40
2	48.58	3.77	7.77	49.77	4.57	9.19
3	47.60	2.04	4.28	54.39	2.13	3.92
4	51.16	1.45	2.83	51.21	1.32	2.58
5	49.42	1.07	2.17	56.58	1.07	1.90

Table 29: Estimates and Standard Errors for Dropping Out by Quintile Classes-Kerala Rural

Quintile Class	Male			Female		
	Estimate (%)	SE (%)	RSE	Estimate (%)	SE (%)	RSE
1	57.39	5.27	9.17	53.22	3.94	7.40
2	48.58	3.77	7.77	49.77	4.57	9.19
3	47.28	2.03	4.30	53.77	2.09	3.89
4	51.06	1.45	2.84	50.89	1.32	2.59
5	48.21	1.05	2.17	55.11	1.04	1.88

Table 30: Estimates and Standard Errors for Non-enrollment by Quintile Classes-India Urban

Quintile Class	Male			Female		
	Estimate (%)	SE (%)	RSE	Estimate (%)	SE (%)	RSE
1	18.85	0.34	1.81	24.42	0.44	1.80
2	13.45	0.24	1.75	16.40	0.27	1.62
3	9.19	0.16	1.78	9.64	0.18	1.88
4	6.17	0.11	1.85	8.33	0.18	2.19
5	3.18	0.11	3.39	5.33	0.31	5.76

Table 31: Estimates and Standard Errors for Ever Enrolled but Currently not attending by Quintile Classes-India Urban

Quintile Class	Male			Female		
	Estimate (%)	SE (%)	RSE	Estimate (%)	SE (%)	RSE
1	49.35	0.68	1.38	45.70	0.69	1.52
2	52.03	0.71	1.36	53.69	0.73	1.36
3	58.46	0.69	1.18	59.15	0.70	1.18
4	61.15	0.71	1.16	58.18	0.68	1.17
5	58.04	0.64	1.11	62.12	0.86	1.39

Table 32: Estimates and Standard Errors for Dropping Out by Quintile Classes—India Urban

Quintile Class	Male			Female		
	Estimate (%)	SE (%)	RSE	Estimate (%)	SE (%)	RSE
1	48.87	0.68	1.39	45.32	0.69	1.52
2	51.41	0.70	1.37	53.10	0.72	1.36
3	57.26	0.67	1.18	58.30	0.69	1.19
4	59.36	0.69	1.17	56.67	0.66	1.17
5	55.27	0.63	1.13	60.38	0.84	1.39

Table 33: Estimates and Standard Errors for Non-enrollment by Quintile Classes—Kerala Urban

Quintile Class	Male			Female		
	Estimate (%)	SE (%)	RSE	Estimate (%)	SE (%)	RSE
1	9.39	0.55	5.87	1.92	0.31	16.04
2	5.96	0.52	8.71	6.87	0.47	6.78
3	7.39	0.54	7.37	2.63	0.10	3.77
4	2.77	0.08	2.89	2.47	0.58	23.51
5	2.05	0.48	23.37	3.49	0.28	8.00

Table 34: Estimates and Standard Errors for Ever Enrolled but Currently not attending by Quintile Classes—Kerala Urban

Quintile Class	Male			Female		
	Estimate (%)	SE (%)	RSE	Estimate (%)	SE (%)	RSE
1	54.29	2.31	4.26	58.96	2.58	4.37
2	42.72	1.39	3.26	59.58	1.86	3.13
3	55.73	1.54	2.76	53.13	1.38	2.59
4	49.80	1.48	2.98	51.28	1.20	2.34
5	51.36	1.62	3.16	54.81	1.97	3.60

Table 35: Estimates and Standard Errors for Dropping Out by Quintile Classes—Kerala Urban

Quintile Class	Male			Female		
	Estimate (%)	SE (%)	RSE	Estimate (%)	SE (%)	RSE
1	51.29	2.25	4.38	58.45	2.57	4.39
2	42.72	1.39	3.26	59.24	1.86	3.14
3	52.76	1.52	2.88	52.31	1.37	2.62
4	49.20	1.49	3.02	50.98	1.19	2.34
5	49.41	1.60	3.25	52.33	1.92	3.66

Note: The percentages are calculated for each quintile class gender-wise and sector-wise for ages between 3-35 years

5 Conclusion

1. We observe that Kerala has a significantly lower percentage of people who have never enrolled for education compared to all of India. The percentage of people from rural areas who have never enrolled in education is higher than the percentage of people who have never enrolled in education from urban areas for all of India. However, in Kerala these percentages are almost similar, indicating balanced human development in Kerala across sectors compared to all of India. The percentage of people who have dropped out is very low in Kerala, compared to the rest of India, whereas the percentage of people ever enrolled but not currently attending education is almost similar to the all-India level. This indicates that more people stop attending education only after reaching their desired qualification in Kerala, compared to the rest of India.
2. Across India, the predominant reason for discontinuation of education and dropping out is engagement in economic activities for males and engagement in domestic activities for females. However, in Kerala, the major reason for discontinuation of education in both males and females in urban areas is the inability to cope with academics/failure in studies. In rural areas, for males the major reason for discontinuation is either a lack of interest in education or the completion of the desired level of education. For females, the primary reason for discontinuation of studies is due to marriage.
3. Across quintile classes, the percentage of non-enrollment is much less in Kerala compared to the all-India level. The percentage of non-enrollment in urban areas is much less compared to that in rural areas across quintile classes all over India.

6 R Codes

All the estimation was done based on NSSO 75th Round Data using R. Below is the entire set of code used for computation:

```
# =====  
# Educational Enrollment Analysis - NSS Survey Data  
# =====  
# This script analyzes educational enrollment patterns across India and Kerala  
# using NSS (National Sample Survey) data, implementing ratio estimation with  
# variance calculation for complex survey designs.  
# =====  
  
# -----  
# 1. LOAD REQUIRED LIBRARIES  
# -----  
  
library(reshape2)  
library(dplyr)  
library(tidyr)  
library(haven)  
  
# -----  
# 2. DEFINE ESTIMATION FUNCTIONS  
# -----  
  
# Calculate weighted total for merged data (with .x suffix)  
wt_tot_x <- function(data) {  
  return(sum(data$MULT_Combined.x, na.rm = TRUE))  
}  
  
# Calculate ratio estimate for merged data  
Rhat_function_x <- function(data) {  
  R <- (wt_tot_x(data[which(data$yind == 1), ])/  
        wt_tot(data[which(data$xind == 1), ]))  
  return(R)  
}  
  
# Calculate variance estimate for merged data  
varest_x <- function(data) {  
  R <- Rhat_function(data)  
  data_split <- split(data, data$Stratum.x)  
  var <- 0
```

```

for (s in 1:length(data_split)) {
  vars <- 0
  data_s <- data_split[[s]]
  data_s_split <- split(data_s, data_s$Sub_stratum.x)

  for (t in 1:length(data_s_split)) {
    varst <- 0
    data_st <- data_s_split[[t]]
    data_st_split <- split(data_st, data_st$Sub_sample.x)

    if (length(data_st_split) < 1) next

    yest1 <- wt_tot(data_st_split[[1]][which(data_st_split[[1]]$yind == 1), ])
    xest1 <- wt_tot(data_st_split[[1]][which(data_st_split[[1]]$xind == 1), ])

    if (length(data_st_split) < 2) {
      yest2 <- xest2 <- 0
    } else {
      yest2 <- wt_tot(data_st_split[[2]][which(data_st_split[[2]]$yind == 1), ])
      xest2 <- wt_tot(data_st_split[[2]][which(data_st_split[[2]]$xind == 1), ])
    }

    varst <- (yest1 - yest2)^2 + (R^2) * ((xest1 - xest2)^2) -
      2 * R * (yest1 - yest2) * (xest1 - xest2)
    vars <- vars + varst
  }
  var <- var + vars
}
return(var / (4 * wt_tot(data[which(xind == 1), ])^2))
}

# Calculate weighted total for standard data
wt_tot <- function(data) {
  return(sum(data$MULT_Combined, na.rm = TRUE))
}

# Calculate ratio estimate for standard data
Rhat_function <- function(data) {
  R <- (wt_tot(data[which(data$yind == 1), ]) /
    wt_tot(data[which(data$xind == 1), ]))
  return(R)
}

# Calculate variance estimate for standard data using stratified two-stage design
varest <- function(data) {
  R <- Rhat_function(data)
  data_split <- split(data, data$Stratum)
  var <- 0

  for (s in 1:length(data_split)) {
    vars <- 0
    data_s <- data_split[[s]]
    data_s_split <- split(data_s, data_s$Sub_stratum)

    for (t in 1:length(data_s_split)) {
      varst <- 0
      data_st <- data_s_split[[t]]
      data_st_split <- split(data_st, data_st$Sub_sample)

      if (length(data_st_split) < 1) next

      yest1 <- wt_tot(data_st_split[[1]][which(data_st_split[[1]]$yind == 1), ])
      xest1 <- wt_tot(data_st_split[[1]][which(data_st_split[[1]]$xind == 1), ])

      if (length(data_st_split) < 2) {
        yest2 <- xest2 <- 0
      } else {
        yest2 <- wt_tot(data_st_split[[2]][which(data_st_split[[2]]$yind == 1), ])
        xest2 <- wt_tot(data_st_split[[2]][which(data_st_split[[2]]$xind == 1), ])
      }

      varst <- (yest1 - yest2)^2 + (R^2) * ((xest1 - xest2)^2) -
        2 * R * (yest1 - yest2) * (xest1 - xest2)
      vars <- vars + varst
    }
    var <- var + vars
  }
  return(var / (4 * wt_tot(data[which(xind == 1), ])^2))
}

# -----

```

```

# 3. DATA LOADING AND PREPROCESSING
# -----

setwd("D:\\ISI\\Semester 5 B.Stat\\Sample Surveys\\SS Assignment 2")

# Load survey blocks
block3 <- read_sav("Block 3.sav")
block4 <- read_sav("Block 4.sav")
block7 <- read_sav("Block 7.sav")

# Clean Block 4 data
block4 <- block4 %>%
  zap_labels() %>%
  zap_formats() %>%
  zap_label() %>%
  mutate(across(where(is.character), ~ trimws(.x))) %>%
  mutate(across(where(~ all(grepl("[0-9]+$", .x)) & is.character(.x)), as.numeric)) %>%
  mutate(
    Age = as.numeric(trimws(Age)),
    enrolmt_3_35_yrs_status = as.numeric(trimws(enrolmt_3_35_yrs_status)),
    MULT_Combined = as.numeric(trimws(MULT_Combined))
  )

# Extract Kerala data (State code = 32)
s <- "32"
block7ker <- subset(block7, State == s)
block4ker <- subset(block4, State == s)
block3ker <- subset(block3, State == s)

# Clean Block 3 data (India and Kerala)
block3 <- block3 %>%
  zap_labels() %>%
  mutate(
    HH_Con_exp_rs = as.numeric(trimws(HH_Con_exp_rs)),
    Household_size = as.numeric(trimws(Household_size)),
    MULT_Combined = as.numeric(trimws(MULT_Combined))
  )

block3ker <- block3ker %>%
  zap_labels() %>%
  mutate(
    HH_Con_exp_rs = as.numeric(trimws(HH_Con_exp_rs)),
    Household_size = as.numeric(trimws(Household_size)),
    MULT_Combined = as.numeric(trimws(MULT_Combined))
  )

# Clean Kerala Block 4 data
block4ker <- block4ker %>%
  zap_labels() %>%
  mutate(across(where(is.character), ~ trimws(.x))) %>%
  mutate(across(where(~ all(grepl("[0-9]+$", .x)) & is.character(.x)), as.numeric)) %>%
  mutate(
    Age = as.numeric(trimws(Age)),
    enrolmt_3_35_yrs_status = as.numeric(trimws(enrolmt_3_35_yrs_status)),
    MULT_Combined = as.numeric(trimws(MULT_Combined))
  )

# -----
# 4. ANALYSIS 1: NEVER ENROLLMENT RATES
# -----

## 4.1 India - Never Enrollment by Sector and Gender
india_est_ne <- data.frame(sector = c(), sex = c(), r = c(), se = c(), rse = c())

for (sec in 1:2) {
  for (gen in 1:2) {
    xind <- ifelse(block4$Sector == sec & block4$Gender == gen &
      block4$Age >= 3 & block4$Age <= 35, 1, 0)
    yind <- ifelse(block4$Sector == sec & block4$Gender == gen &
      block4$enrolmt_3_35_yrs_status == 1, 1, 0)
    temp <- cbind(block4, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100
    india_est_ne <- rbind(india_est_ne,
      data.frame(sector = sec, sex = gen, r = R, se = se, rse = rse))
  }
}
print(india_est_ne)

## 4.2 Kerala - Never Enrollment by Sector and Gender

```

```

ker_est_ne <- data.frame(sector = c(), sex = c(), r = c(), se = c(), rse = c())

for (sec in 1:2) {
  for (gen in 1:2) {
    xind <- ifelse(block4ker$Sector == sec & block4ker$Gender == gen &
      block4ker$Age >= 3 & block4ker$Age <= 35, 1, 0)
    yind <- ifelse(block4ker$Sector == sec & block4ker$Gender == gen &
      block4ker$enrolmt_3_35_yrs_status == 1, 1, 0)
    temp <- cbind(block4ker, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100
    ker_est_ne <- rbind(ker_est_ne,
      data.frame(sector = sec, sex = gen, r = R, se = se, rse = rse))
  }
}
print(ker_est_ne)

# -----
# 5. ANALYSIS 2: DROPOUT RATES
# -----

## 5.1 India - Dropout Rates
colat_ind <- merge(block4, block7, by = c("HHID", "Per_serialno"), all.x = TRUE)
colat_ind <- colat_ind %>%
  zap_labels() %>%
  mutate(across(where(is.character), trimws)) %>%
  mutate(across(where(~ all(grepl("^-[0-9.]*$", .x[!is.na(.x)]))), as.numeric))

ind_est_do <- data.frame(sector = c(), sex = c(), r = c(), se = c(), rse = c())

for (sec in 1:2) {
  for (gen in 1:2) {
    xind <- ifelse(colat_ind$Sector.x == sec & colat_ind$Gender == gen &
      colat_ind$Age.x >= 3 & colat_ind$Age.x <= 35, 1, 0)
    yind <- ifelse(colat_ind$Sector.x == sec & colat_ind$Gender == gen &
      colat_ind$Whether_ever_attended == 1 &
      colat_ind$Completed_last_enrolled == 2 &
      colat_ind$Major_reason_not_enrolled != 15, 1, 0)
    temp <- cbind(colat_ind, xind, yind)
    R <- Rhat_function_x(temp) * 100
    se <- sqrt(varest_x(temp)) * 100
    rse <- se / R * 100
    ind_est_do <- rbind(ind_est_do,
      data.frame(sector = sec, sex = gen, r = R, se = se, rse = rse))
  }
}
print(ind_est_do)

## 5.2 Kerala - Dropout Rates
colat_ker <- subset(colat_ind, State.x == "32")
ker_est_do <- data.frame(sector = c(), sex = c(), r = c(), se = c(), rse = c())

for (sec in 1:2) {
  for (gen in 1:2) {
    xind <- ifelse(colat_ker$Sector.x == sec & colat_ker$Gender == gen &
      colat_ker$Age.x >= 3 & colat_ker$Age.x <= 35, 1, 0)
    yind <- ifelse(colat_ker$Sector.x == sec & colat_ker$Gender == gen &
      colat_ker$Whether_ever_attended == 1 &
      colat_ker$Completed_last_enrolled == 2 &
      colat_ker$Major_reason_not_enrolled != 15, 1, 0)
    temp <- cbind(colat_ker, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100
    ker_est_do <- rbind(ker_est_do,
      data.frame(sector = sec, sex = gen, r = R, se = se, rse = rse))
  }
}
print(ker_est_do)

# -----
# 6. ANALYSIS 3: EVER ENROLLED BUT CURRENTLY NOT ATTENDING
# -----

## 6.1 India - Not Currently Attending
india_est_na <- data.frame(sector = c(), sex = c(), r = c(), se = c(), rse = c())

for (sec in 1:2) {
  for (gen in 1:2) {
    xind <- ifelse(block4$Sector == sec & block4$Gender == gen &

```

```

        block4$Age >= 3 & block4$Age <= 35, 1, 0)
y1ind <- ifelse(block4$Sector == sec & block4$Gender == gen &
        block4$enrolmt_3_35_yrs_status == 2, 1, 0)
y2ind <- ifelse(block4$Sector == sec & block4$Gender == gen &
        block4$enrolmt_3_35_yrs_status == 3, 1, 0)
yind <- (y1ind + y2ind) %% 2

temp <- cbind(block4, xind, yind)
R <- Rhat_function(temp) * 100
se <- sqrt(varest(temp)) * 100
rse <- se / R * 100
india_est_na <- rbind(india_est_na,
        data.frame(sector = sec, sex = gen, r = R, se = se, rse = rse))
}
}
print(india_est_na)

## 6.2 Kerala - Not Currently Attending
ker_est_na <- data.frame(sector = c(), sex = c(), r = c(), se = c(), rse = c())

for (sec in 1:2) {
  for (gen in 1:2) {
    xind <- ifelse(block4ker$Sector == sec & block4ker$Gender == gen &
        block4ker$Age >= 3 & block4ker$Age <= 35, 1, 0)
    yind1 <- ifelse(block4ker$Sector == sec & block4ker$Gender == gen &
        block4ker$enrolmt_3_35_yrs_status == 2, 1, 0)
    yind2 <- ifelse(block4ker$Sector == sec & block4ker$Gender == gen &
        block4ker$enrolmt_3_35_yrs_status == 3, 1, 0)
    yind <- (yind1 + yind2) %% 2

    temp <- cbind(block4ker, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100
    ker_est_na <- rbind(ker_est_na,
        data.frame(sector = sec, sex = gen, r = R, se = se, rse = rse))
  }
}
print(ker_est_na)

# -----
# 7. ANALYSIS 4: REASONS FOR NON-ENROLLMENT AND NON-ATTENDANCE
# -----

# Merge enrollment status with reasons data
data <- merge(block7,
        block4[, c("HHID", "Per_serialno", "Gender", "enrolmt_3_35_yrs_status")],
        by = c("HHID", "Per_serialno"), all.x = TRUE)
data <- data %>%
  zap_labels() %>%
  mutate(across(where(is.character), trimws)) %>%
  mutate(across(where(~ all(grepl("^-[0-9.]*$", .x[!is.na(.x)]))), as.numeric))

# Reason codes (excluding code 9)
vec <- c(1:8, 10:20)

## 7.1 India - Reasons for Never Enrollment (Males)
india_reason_ne_male <- data.frame(sector = c(), reason = c(), r = c(),
        se = c(), rse = c())

for (sec in 1:2) {
  for (reas in vec) {
    xind <- ifelse(as.numeric(data$Sector) == sec & data$Gender == 1 &
        data$enrolmt_3_35_yrs_status == 1, 1, 0)
    yind <- ifelse(as.numeric(data$Sector) == sec & data$Gender == 1 &
        data$enrolmt_3_35_yrs_status == 1 &
        as.numeric(data$Major_reason_not_enrolled) == reas, 1, 0)
    temp <- cbind(data, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100.0
    india_reason_ne_male <- rbind(india_reason_ne_male,
        data.frame(sector = sec, reason = reas,
            r = R, se = se, rse = rse))
  }
}
print(india_reason_ne_male)

## 7.2 India - Reasons for Not Attending (Males)
india_reason_na_male <- data.frame(sector = c(), reason = c(), r = c(),
        se = c(), rse = c())

```

```

for (sec in 1:2) {
  for (reas in vec) {
    xind1 <- ifelse(as.numeric(data$Sector) == sec & data$Gender == 1 &
      data$enrolmt_3_35_yrs_status == 2, 1, 0)
    xind2 <- ifelse(as.numeric(data$Sector) == sec & data$Gender == 1 &
      data$enrolmt_3_35_yrs_status == 3, 1, 0)
    xind <- xind1 + xind2

    yind1 <- ifelse(as.numeric(data$Sector) == sec & data$Gender == 1 &
      data$enrolmt_3_35_yrs_status == 2 &
      as.numeric(data$Major_reason_not_enrolled) == reas, 1, 0)
    yind2 <- ifelse(as.numeric(data$Sector) == sec & data$Gender == 1 &
      data$enrolmt_3_35_yrs_status == 3 &
      as.numeric(data$Major_reason_not_enrolled) == reas, 1, 0)
    yind <- yind1 + yind2

    temp <- cbind(data, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100.0
    india_reason_na_male <- rbind(india_reason_na_male,
      data.frame(sector = sec, reason = reas,
        r = R, se = se, rse = rse))
  }
}
print(india_reason_na_male)

## 7.3 India - Reasons for Dropout (Males)
india_reason_do_male <- data.frame(sector = c(), reason = c(), r = c(),
  se = c(), rse = c())

for (sec in 1:2) {
  for (reas in vec) {
    xind1 <- ifelse(as.numeric(data$Sector) == sec & data$Gender == 1 &
      data$Major_reason_not_enrolled != 15 &
      data$Completed_last_enrolled == 2 &
      data$enrolmt_3_35_yrs_status == 2, 1, 0)
    xind2 <- ifelse(as.numeric(data$Sector) == sec & data$Gender == 1 &
      data$Major_reason_not_enrolled != 15 &
      data$Completed_last_enrolled == 2 &
      data$enrolmt_3_35_yrs_status == 3, 1, 0)
    xind <- xind1 + xind2

    yind1 <- ifelse(as.numeric(data$Sector) == sec & data$Gender == 1 &
      data$Major_reason_not_enrolled != 15 &
      data$Completed_last_enrolled == 2 &
      data$enrolmt_3_35_yrs_status == 2 &
      as.numeric(data$Major_reason_not_enrolled) == reas, 1, 0)
    yind2 <- ifelse(as.numeric(data$Sector) == sec & data$Gender == 1 &
      data$Major_reason_not_enrolled != 15 &
      data$Completed_last_enrolled == 2 &
      data$enrolmt_3_35_yrs_status == 3 &
      as.numeric(data$Major_reason_not_enrolled) == reas, 1, 0)
    yind <- yind1 + yind2

    temp <- cbind(data, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100.0
    india_reason_do_male <- rbind(india_reason_do_male,
      data.frame(sector = sec, reason = reas,
        r = R, se = se, rse = rse))
  }
}
print(india_reason_do_male)

## 7.4 India - Reasons for Never Enrollment (Females)
india_reason_ne_female <- data.frame(sector = c(), reason = c(), r = c(),
  se = c(), rse = c())

for (sec in 1:2) {
  for (reas in vec) {
    xind <- ifelse(as.numeric(data$Sector) == sec & data$Gender == 2 &
      data$enrolmt_3_35_yrs_status == 1, 1, 0)
    yind <- ifelse(as.numeric(data$Sector) == sec & data$Gender == 2 &
      data$enrolmt_3_35_yrs_status == 1 &
      as.numeric(data$Major_reason_not_enrolled) == reas, 1, 0)
    temp <- cbind(data, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100

```

```

rse <- se / R * 100.0
india_reason_ne_female <- rbind(india_reason_ne_female,
                                data.frame(sector = sec, reason = reas,
                                             r = R, se = se, rse = rse))
}
}
print(india_reason_ne_female)

## 7.5 India - Reasons for Not Attending (Females)
india_reason_na_female <- data.frame(sector = c(), reason = c(), r = c(),
                                     se = c(), rse = c())

for (sec in 1:2) {
  for (reas in vec) {
    xind1 <- ifelse(as.numeric(data$Sector) == sec & data$Gender == 2 &
                    data$enrolmt_3_35_yrs_status == 2, 1, 0)
    xind2 <- ifelse(as.numeric(data$Sector) == sec & data$Gender == 2 &
                    data$enrolmt_3_35_yrs_status == 3, 1, 0)
    xind <- xind1 + xind2

    yind1 <- ifelse(as.numeric(data$Sector) == sec & data$Gender == 2 &
                    data$enrolmt_3_35_yrs_status == 2 &
                    as.numeric(data$Major_reason_not_enrolled) == reas, 1, 0)
    yind2 <- ifelse(as.numeric(data$Sector) == sec & data$Gender == 2 &
                    data$enrolmt_3_35_yrs_status == 3 &
                    as.numeric(data$Major_reason_not_enrolled) == reas, 1, 0)
    yind <- yind1 + yind2

    temp <- cbind(data, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100.0
    india_reason_na_female <- rbind(india_reason_na_female,
                                    data.frame(sector = sec, reason = reas,
                                                 r = R, se = se, rse = rse))
  }
}
print(india_reason_na_female)

## 7.6 India - Reasons for Dropout (Females)
india_reason_do_female <- data.frame(sector = c(), reason = c(), r = c(),
                                     se = c(), rse = c())

for (sec in 1:2) {
  for (reas in vec) {
    xind1 <- ifelse(as.numeric(data$Sector) == sec & data$Gender == 2 &
                    data$Major_reason_not_enrolled != 15 &
                    data$Completed_last_enrolled == 2 &
                    data$enrolmt_3_35_yrs_status == 2, 1, 0)
    xind2 <- ifelse(as.numeric(data$Sector) == sec & data$Gender == 2 &
                    data$Major_reason_not_enrolled != 15 &
                    data$Completed_last_enrolled == 2 &
                    data$enrolmt_3_35_yrs_status == 3, 1, 0)
    xind <- xind1 + xind2

    yind1 <- ifelse(as.numeric(data$Sector) == sec & data$Gender == 2 &
                    data$Major_reason_not_enrolled != 15 &
                    data$Completed_last_enrolled == 2 &
                    data$enrolmt_3_35_yrs_status == 2 &
                    as.numeric(data$Major_reason_not_enrolled) == reas, 1, 0)
    yind2 <- ifelse(as.numeric(data$Sector) == sec & data$Gender == 2 &
                    data$Major_reason_not_enrolled != 15 &
                    data$Completed_last_enrolled == 2 &
                    data$enrolmt_3_35_yrs_status == 3 &
                    as.numeric(data$Major_reason_not_enrolled) == reas, 1, 0)
    yind <- yind1 + yind2

    temp <- cbind(data, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100.0
    india_reason_do_female <- rbind(india_reason_do_female,
                                    data.frame(sector = sec, reason = reas,
                                                 r = R, se = se, rse = rse))
  }
}
print(india_reason_do_female)

# -----
# 8. KERALA-SPECIFIC REASONS ANALYSIS
# -----

```

```

# Prepare Kerala-specific merged data
data_ker <- merge(block7ker,
  block4ker[, c("HHID", "Per_serialno", "Gender", "enrolmt_3_35_yrs_status")],
  by = c("HHID", "Per_serialno"), all.x = TRUE)
data_ker <- data_ker %>%
  zap_labels() %>%
  mutate(across(where(is.character), trimws)) %>%
  mutate(across(where(~ all(grepl("^-[0-9.]*$", .x[!is.na(.x)]))), as.numeric))

## 8.1 Kerala - Reasons for Never Enrollment (Males)
ker_reason_ne_male <- data.frame(sector = c(), reason = c(), r = c(),
  se = c(), rse = c())

for (sec in 1:2) {
  for (reas in vec) {
    xind <- ifelse(as.numeric(data_ker$Sector) == sec & data_ker$Gender == 1 &
      data_ker$enrolmt_3_35_yrs_status == 1, 1, 0)
    yind <- ifelse(as.numeric(data_ker$Sector) == sec & data_ker$Gender == 1 &
      data_ker$enrolmt_3_35_yrs_status == 1 &
      as.numeric(data_ker$Major_reason_not_enrolled) == reas, 1, 0)
    temp <- cbind(data_ker, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100.0
    ker_reason_ne_male <- rbind(ker_reason_ne_male,
      data.frame(sector = sec, reason = reas,
        r = R, se = se, rse = rse))
  }
}
print(ker_reason_ne_male)

## 8.2 Kerala - Reasons for Not Attending (Males)
ker_reason_na_male <- data.frame(sector = c(), reason = c(), r = c(),
  se = c(), rse = c())

for (sec in 1:2) {
  for (reas in vec) {
    xind1 <- ifelse(as.numeric(data_ker$Sector) == sec & data_ker$Gender == 1 &
      data_ker$enrolmt_3_35_yrs_status == 2, 1, 0)
    xind2 <- ifelse(as.numeric(data_ker$Sector) == sec & data_ker$Gender == 1 &
      data_ker$enrolmt_3_35_yrs_status == 3, 1, 0)
    xind <- xind1 + xind2

    yind1 <- ifelse(as.numeric(data_ker$Sector) == sec & data_ker$Gender == 1 &
      data_ker$enrolmt_3_35_yrs_status == 2 &
      as.numeric(data_ker$Major_reason_not_enrolled) == reas, 1, 0)
    yind2 <- ifelse(as.numeric(data_ker$Sector) == sec & data_ker$Gender == 1 &
      data_ker$enrolmt_3_35_yrs_status == 3 &
      as.numeric(data_ker$Major_reason_not_enrolled) == reas, 1, 0)
    yind <- yind1 + yind2

    temp <- cbind(data_ker, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100.0
    ker_reason_na_male <- rbind(ker_reason_na_male,
      data.frame(sector = sec, reason = reas,
        r = R, se = se, rse = rse))
  }
}
print(ker_reason_na_male)

## 8.3 Kerala - Reasons for Dropout (Males)
ker_reason_do_male <- data.frame(sector = c(), reason = c(), r = c(),
  se = c(), rse = c())

for (sec in 1:2) {
  for (reas in vec) {
    xind1 <- ifelse(as.numeric(data_ker$Sector) == sec & data_ker$Gender == 1 &
      data_ker$Major_reason_not_enrolled != 15 &
      data_ker$Completed_last_enrolled == 2 &
      data_ker$enrolmt_3_35_yrs_status == 2, 1, 0)
    xind2 <- ifelse(as.numeric(data_ker$Sector) == sec & data_ker$Gender == 1 &
      data_ker$Major_reason_not_enrolled != 15 &
      data_ker$Completed_last_enrolled == 2 &
      data_ker$enrolmt_3_35_yrs_status == 3, 1, 0)
    xind <- xind1 + xind2

    yind1 <- ifelse(as.numeric(data_ker$Sector) == sec & data_ker$Gender == 1 &
      data_ker$Major_reason_not_enrolled != 15 &

```

```

        data_ker$Completed_last_enrolled == 2 &
        data_ker$enrolmt_3_35_yrs_status == 2 &
        as.numeric(data_ker$Major_reason_not_enrolled) == reas, 1, 0)
yind2 <- ifelse(as.numeric(data_ker$Sector) == sec & data_ker$Gender == 1 &
        data_ker$Major_reason_not_enrolled != 15 &
        data_ker$Completed_last_enrolled == 2 &
        data_ker$enrolmt_3_35_yrs_status == 3 &
        as.numeric(data_ker$Major_reason_not_enrolled) == reas, 1, 0)
yind <- yind1 + yind2

temp <- cbind(data_ker, xind, yind)
R <- Rhat_function(temp) * 100
se <- sqrt(varest(temp)) * 100
rse <- se / R * 100.0
ker_reason_do_male <- rbind(ker_reason_do_male,
        data.frame(sector = sec, reason = reas,
                r = R, se = se, rse = rse))
}
}
print(ker_reason_do_male)

## 8.4 Kerala - Reasons for Never Enrollment (Females)
ker_reason_ne_female <- data.frame(sector = c(), reason = c(), r = c(),
        se = c(), rse = c())

for (sec in 1:2) {
  for (reas in vec) {
    xind <- ifelse(as.numeric(data_ker$Sector) == sec & data_ker$Gender == 2 &
        data_ker$enrolmt_3_35_yrs_status == 1, 1, 0)
    yind <- ifelse(as.numeric(data_ker$Sector) == sec & data_ker$Gender == 2 &
        data_ker$enrolmt_3_35_yrs_status == 1 &
        as.numeric(data_ker$Major_reason_not_enrolled) == reas, 1, 0)
    temp <- cbind(data_ker, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100.0
    ker_reason_ne_female <- rbind(ker_reason_ne_female,
            data.frame(sector = sec, reason = reas,
                    r = R, se = se, rse = rse))
  }
}
print(ker_reason_ne_female)

## 8.5 Kerala - Reasons for Not Attending (Females)
ker_reason_na_female <- data.frame(sector = c(), reason = c(), r = c(),
        se = c(), rse = c())

for (sec in 1:2) {
  for (reas in vec) {
    xind1 <- ifelse(as.numeric(data_ker$Sector) == sec & data_ker$Gender == 2 &
        data_ker$enrolmt_3_35_yrs_status == 2, 1, 0)
    xind2 <- ifelse(as.numeric(data_ker$Sector) == sec & data_ker$Gender == 2 &
        data_ker$enrolmt_3_35_yrs_status == 3, 1, 0)
    xind <- xind1 + xind2

    yind1 <- ifelse(as.numeric(data_ker$Sector) == sec & data_ker$Gender == 2 &
        data_ker$enrolmt_3_35_yrs_status == 2 &
        as.numeric(data_ker$Major_reason_not_enrolled) == reas, 1, 0)
    yind2 <- ifelse(as.numeric(data_ker$Sector) == sec & data_ker$Gender == 2 &
        data_ker$enrolmt_3_35_yrs_status == 3 &
        as.numeric(data_ker$Major_reason_not_enrolled) == reas, 1, 0)
    yind <- yind1 + yind2

    temp <- cbind(data_ker, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100.0
    ker_reason_na_female <- rbind(ker_reason_na_female,
            data.frame(sector = sec, reason = reas,
                    r = R, se = se, rse = rse))
  }
}
print(ker_reason_na_female)

## 8.6 Kerala - Reasons for Dropout (Females)
ker_reason_do_female <- data.frame(sector = c(), reason = c(), r = c(),
        se = c(), rse = c())

for (sec in 1:2) {
  for (reas in vec) {
    xind1 <- ifelse(as.numeric(data_ker$Sector) == sec & data_ker$Gender == 2 &

```

```

        data_ker$Major_reason_not_enrolled != 15 &
        data_ker$Completed_last_enrolled == 2 &
        data_ker$enrolmt_3_35_yrs_status == 2, 1, 0)
xind2 <- ifelse(as.numeric(data_ker$Sector) == sec & data_ker$Gender == 2 &
        data_ker$Major_reason_not_enrolled != 15 &
        data_ker$Completed_last_enrolled == 2 &
        data_ker$enrolmt_3_35_yrs_status == 3, 1, 0)
xind <- xind1 + xind2

yind1 <- ifelse(as.numeric(data_ker$Sector) == sec & data_ker$Gender == 2 &
        data_ker$Major_reason_not_enrolled != 15 &
        data_ker$Completed_last_enrolled == 2 &
        data_ker$enrolmt_3_35_yrs_status == 2 &
        as.numeric(data_ker$Major_reason_not_enrolled) == reas, 1, 0)
yind2 <- ifelse(as.numeric(data_ker$Sector) == sec & data_ker$Gender == 2 &
        data_ker$Major_reason_not_enrolled != 15 &
        data_ker$Completed_last_enrolled == 2 &
        data_ker$enrolmt_3_35_yrs_status == 3 &
        as.numeric(data_ker$Major_reason_not_enrolled) == reas, 1, 0)
yind <- yind1 + yind2

temp <- cbind(data_ker, xind, yind)
R <- Rhat_function(temp) * 100
se <- sqrt(varest(temp)) * 100
rse <- se / R * 100.0
ker_reason_do_female <- rbind(ker_reason_do_female,
        data.frame(sectors = sec, reasons = reas,
                r = R, se = se, rse = rse))
}
}
print(ker_reason_do_female)

# -----
# 9. UMPCE QUINTILE CALCULATION
# -----

# Calculate UMPCE (Uniform Monthly Per Capita Expenditure) quintiles
perc_seq <- seq(0.2, 0.8, by = 0.2)
quintiles <- data.frame(group = c(), sector = c(), k1 = c(), k2 = c(),
        k3 = c(), k4 = c())

for (i in 1:2) {
  for (sec in 1:2) {
    cutoffs <- c()

    if (i == 1) {
      data <- subset(block3, Sector == sec)
    } else {
      data <- subset(block3ker, Sector == sec)
    }

    umpce <- data$HH_Con_exp_rs / data$Household_size
    est_pop_per_house <- data$Household_size * data$MULT_Combined
    temp <- cbind(data, umpce, est_pop_per_house)
    total_pop <- sum(est_pop_per_house)
    temp <- temp %>% arrange(umpce)
    cumulative_perc <- cumsum(temp$est_pop_per_house) / total_pop

    for (k in 1:4) {
      ser <- (1:length(cumulative_perc))[cumulative_perc <= perc_seq[k]]
      cutoffs <- c(cutoffs, temp$umpce[max(ser)])
    }

    quintiles <- rbind(quintiles,
            data.frame(group = i, Sector = sec, k1 = cutoffs[1],
                    k2 = cutoffs[2], k3 = cutoffs[3], k4 = cutoffs[4]))
  }
}
print(quintiles)

# -----
# 10. QUINTILE-BASED ANALYSIS
# -----

# Prepare data with UMPCE quintile classification
data <- block3[!is.na(block3$HH_Con_exp_rs), ]
umpce <- data$HH_Con_exp_rs / data$Household_size
est_pop_per_sample <- data$Household_size * data$MULT_Combined
data <- data.frame(cbind(data, umpce, est_pop_per_sample))
pseq <- c(0.2, 0.4, 0.6, 0.8)

```

```

# Calculate quintiles for each sector and assign classes
for (i in 1:2) {
  temp <- subset(data, Sector == i)
  temp_sorted <- data.frame(temp[order(temp$umpce), ])
  cum_est_pop <- cumsum(temp_sorted$est_pop_per_sample)
  tot_pop <- sum(temp_sorted$Household_size * temp_sorted$MULT_Combined)
  percent_cum_est_pop <- cum_est_pop / tot_pop
  quint <- c()

  for (j in 1:4) {
    ser <- (1:length(cum_est_pop))[percent_cum_est_pop <= pseq[j]]
    quint[j] <- temp_sorted$umpce[max(ser)]
  }

  cat(i, quint, "\n")
  class <- as.factor(ifelse(temp$umpce < quint[1], 1,
    ifelse(temp$umpce < quint[2], 2,
    ifelse(temp$umpce < quint[3], 3,
    ifelse(temp$umpce < quint[4], 4, 5))))))

  if (i == 1) {
    temp1 <- data.frame(cbind(temp, class))
  }
  if (i == 2) {
    temp2 <- data.frame(cbind(temp, class))
  }
}

# -----
# 11. RURAL INDIA - QUINTILE-BASED ENROLLMENT ANALYSIS
# -----

# Merge household data with individual enrollment data
dat1 <- merge(temp1, block4[, c("HHID", "enrolmt_3_35_yrs_status", "Age", "Gender")],
  by = c("HHID"), all.x = TRUE)
dat11 <- merge(dat1, block7[, c("HHID", "Major_reason_not_enrolled")],
  by = c("HHID"), all.x = TRUE)
dat2 <- subset(dat11, Age <= 35 & Age >= 3 & Sector == 1)
dat1_ker <- subset(dat11, State == 32 & Age <= 35 & Age >= 3 & Sector == 1)

## 11.1 Rural India - Never Enrollment by Quintile
res <- data.frame(class = c(), gender = c(), r = c(), se = c(), rse = c())

for (c in 1:5) {
  for (gen in 1:2) {
    xind <- ifelse(dat2$class == c & dat2$Gender == gen, 1, 0)
    yind <- ifelse(dat2$class == c & dat2$Gender == gen &
      dat2$enrolmt_3_35_yrs_status == 1, 1, 0)
    temp <- cbind(dat2, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100.0
    res <- rbind(res, data.frame(class = c, gender = gen, r = R, se = se, rse = rse))
  }
}
print(res)

## 11.2 Rural India - Ever Enrolled but Not Attending by Quintile
res <- data.frame(class = c(), gender = c(), r = c(), se = c(), rse = c())

for (c in 1:5) {
  for (gen in 1:2) {
    xind <- ifelse(dat2$class == c & dat2$Gender == gen, 1, 0)
    yind1 <- ifelse(dat2$class == c & dat2$Gender == gen &
      dat2$enrolmt_3_35_yrs_status == 2, 1, 0)
    yind2 <- ifelse(dat2$class == c & dat2$Gender == gen &
      dat2$enrolmt_3_35_yrs_status == 3, 1, 0)
    yind <- yind1 + yind2
    temp <- cbind(dat2, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100.0
    res <- rbind(res, data.frame(class = c, gender = gen, r = R, se = se, rse = rse))
  }
}
print(res)

## 11.3 Rural India - Dropout by Quintile
res <- data.frame(class = c(), gender = c(), r = c(), se = c(), rse = c())

for (c in 1:5) {

```

```

for (gen in 1:2) {
  xind <- ifelse(dat2$class == c & dat2$Gender == gen, 1, 0)
  yind1 <- ifelse(dat2$class == c & dat2$Gender == gen &
    as.numeric(as_factor(dat2$Major_reason_not_enrolled)) != 15 &
    dat2$enrolmt_3_35_yrs_status == 2, 1, 0)
  yind2 <- ifelse(dat2$class == c & dat2$Gender == gen &
    as.numeric(as_factor(dat2$Major_reason_not_enrolled)) != 15 &
    dat2$enrolmt_3_35_yrs_status == 3, 1, 0)
  yind <- yind1 + yind2
  temp <- cbind(dat2, xind, yind)
  R <- Rhat_function(temp) * 100
  se <- sqrt(varest(temp)) * 100
  rse <- se / R * 100.0
  res <- rbind(res, data.frame(class = c, gender = gen, r = R, se = se, rse = rse))
}
}
print(res)

# -----
# 12. RURAL KERALA - QUINTILE-BASED ENROLLMENT ANALYSIS
# -----

## 12.1 Rural Kerala - Ever Enrolled but Not Attending by Quintile
res <- data.frame(class = c(), gender = c(), r = c(), se = c(), rse = c())

for (c in 1:5) {
  for (gen in 1:2) {
    xind <- ifelse(dat1_ker$class == c & dat1_ker$Gender == gen, 1, 0)
    yind1 <- ifelse(dat1_ker$class == c & dat1_ker$Gender == gen &
      as.numeric(as_factor(dat1_ker$Major_reason_not_enrolled)) != 15 &
      dat1_ker$enrolmt_3_35_yrs_status == 2, 1, 0)
    yind2 <- ifelse(dat1_ker$class == c & dat1_ker$Gender == gen &
      as.numeric(as_factor(dat1_ker$Major_reason_not_enrolled)) != 15 &
      dat1_ker$enrolmt_3_35_yrs_status == 3, 1, 0)
    yind <- (yind1 + yind2) %% 2
    temp <- cbind(dat1_ker, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100.0
    res <- rbind(res, data.frame(class = c, gender = gen, r = R, se = se, rse = rse))
  }
}
print(res)

## 12.2 Rural Kerala - Never Enrollment by Quintile
res <- data.frame(class = c(), gender = c(), r = c(), se = c(), rse = c())

for (c in 1:5) {
  for (gen in 1:2) {
    xind <- ifelse(dat1_ker$class == c & dat1_ker$Gender == gen, 1, 0)
    yind <- ifelse(dat1_ker$class == c & dat1_ker$Gender == gen &
      dat1_ker$enrolmt_3_35_yrs_status == 1, 1, 0)
    temp <- cbind(dat1_ker, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100.0
    res <- rbind(res, data.frame(class = c, gender = gen, r = R, se = se, rse = rse))
  }
}
print(res)

## 12.3 Rural Kerala - Dropout by Quintile
res <- data.frame(class = c(), gender = c(), r = c(), se = c(), rse = c())

for (c in 1:5) {
  for (gen in 1:2) {
    xind <- ifelse(dat1_ker$class == c & dat1_ker$Gender == gen, 1, 0)
    yind1 <- ifelse(dat1_ker$class == c & dat1_ker$Gender == gen &
      as.numeric(as_factor(dat1_ker$Major_reason_not_enrolled)) != 15 &
      dat1_ker$enrolmt_3_35_yrs_status == 2, 1, 0)
    yind2 <- ifelse(dat1_ker$class == c & dat1_ker$Gender == gen &
      as.numeric(as_factor(dat1_ker$Major_reason_not_enrolled)) != 15 &
      dat1_ker$enrolmt_3_35_yrs_status == 3, 1, 0)
    yind <- (yind1 + yind2) %% 2
    temp <- cbind(dat1_ker, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100.0
    res <- rbind(res, data.frame(class = c, gender = gen, r = R, se = se, rse = rse))
  }
}

```

```

}
}
print(res)

# -----
# 13. URBAN INDIA - QUINTILE-BASED ENROLLMENT ANALYSIS
# -----

# Prepare urban data
dat1 <- merge(temp2, block4[, c("HHID", "enrolmt_3_35_yrs_status", "Age", "Gender")],
              by = c("HHID"), all.x = TRUE)
dat11 <- merge(dat1, block7[, c("HHID", "Major_reason_not_enrolled")],
              by = c("HHID"), all.x = TRUE)
dat2 <- subset(dat11, Age <= 35 & Age >= 3 & Sector == 2)
dat1_ker <- subset(dat11, State == 32 & Age <= 35 & Age >= 3 & Sector == 2)

## 13.1 Urban India - Never Enrollment by Quintile
res <- data.frame(class = c(), gender = c(), r = c(), se = c(), rse = c())

for (c in 1:5) {
  for (gen in 1:2) {
    xind <- ifelse(dat2$class == c & dat2$Gender == gen, 1, 0)
    yind <- ifelse(dat2$class == c & dat2$Gender == gen &
                  dat2$enrolmt_3_35_yrs_status == 1, 1, 0)
    temp <- cbind(dat2, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100.0
    res <- rbind(res, data.frame(class = c, gender = gen, r = R, se = se, rse = rse))
  }
}
print(res)

## 13.2 Urban India - Ever Enrolled but Not Attending by Quintile
res <- data.frame(class = c(), gender = c(), r = c(), se = c(), rse = c())

for (c in 1:5) {
  for (gen in 1:2) {
    xind <- ifelse(dat2$class == c & dat2$Gender == gen, 1, 0)
    yind1 <- ifelse(dat2$class == c & dat2$Gender == gen &
                   dat2$enrolmt_3_35_yrs_status == 2, 1, 0)
    yind2 <- ifelse(dat2$class == c & dat2$Gender == gen &
                   dat2$enrolmt_3_35_yrs_status == 3, 1, 0)
    yind <- yind1 + yind2
    temp <- cbind(dat2, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100.0
    res <- rbind(res, data.frame(class = c, gender = gen, r = R, se = se, rse = rse))
  }
}
print(res)

## 13.3 Urban India - Dropout by Quintile
res <- data.frame(class = c(), gender = c(), r = c(), se = c(), rse = c())

for (c in 1:5) {
  for (gen in 1:2) {
    xind <- ifelse(dat2$class == c & dat2$Gender == gen, 1, 0)
    yind1 <- ifelse(dat2$class == c & dat2$Gender == gen &
                   as.numeric(as_factor(dat2$Major_reason_not_enrolled)) != 15 &
                   dat2$enrolmt_3_35_yrs_status == 2, 1, 0)
    yind2 <- ifelse(dat2$class == c & dat2$Gender == gen &
                   as.numeric(as_factor(dat2$Major_reason_not_enrolled)) != 15 &
                   dat2$enrolmt_3_35_yrs_status == 3, 1, 0)
    yind <- yind1 + yind2
    temp <- cbind(dat2, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100.0
    res <- rbind(res, data.frame(class = c, gender = gen, r = R, se = se, rse = rse))
  }
}
print(res)

# -----
# 14. URBAN KERALA - QUINTILE-BASED ENROLLMENT ANALYSIS
# -----

## 14.1 Urban Kerala - Ever Enrolled but Not Attending by Quintile
res <- data.frame(class = c(), gender = c(), r = c(), se = c(), rse = c())

```

```

for (c in 1:5) {
  for (gen in 1:2) {
    xind <- ifelse(dat1_ker$class == c & dat1_ker$Gender == gen, 1, 0)
    yind1 <- ifelse(dat1_ker$class == c & dat1_ker$Gender == gen &
      dat1_ker$enrolmt_3_35_yrs_status == 2, 1, 0)
    yind2 <- ifelse(dat1_ker$class == c & dat1_ker$Gender == gen &
      dat1_ker$enrolmt_3_35_yrs_status == 3, 1, 0)
    yind <- yind1 + yind2
    temp <- cbind(dat1_ker, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100.0
    res <- rbind(res, data.frame(class = c, gender = gen, r = R, se = se, rse = rse))
  }
}
print(res)

## 14.2 Urban Kerala - Never Enrollment by Quintile
res <- data.frame(class = c(), gender = c(), r = c(), se = c(), rse = c())

for (c in 1:5) {
  for (gen in 1:2) {
    xind <- ifelse(dat1_ker$class == c & dat1_ker$Gender == gen, 1, 0)
    yind <- ifelse(dat1_ker$class == c & dat1_ker$Gender == gen &
      dat1_ker$enrolmt_3_35_yrs_status == 1, 1, 0)
    temp <- cbind(dat1_ker, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100.0
    res <- rbind(res, data.frame(class = c, gender = gen, r = R, se = se, rse = rse))
  }
}
print(res)

## 14.3 Urban Kerala - Dropout by Quintile
res <- data.frame(class = c(), gender = c(), r = c(), se = c(), rse = c())

for (c in 1:5) {
  for (gen in 1:2) {
    xind <- ifelse(dat1_ker$class == c & dat1_ker$Gender == gen, 1, 0)
    yind1 <- ifelse(dat1_ker$class == c & dat1_ker$Gender == gen, 1, 0)
    yind1 <- ifelse(dat1_ker$class == c & dat1_ker$Gender == gen &
      dat1_ker$enrolmt_3_35_yrs_status == 2, 1, 0)
    yind2 <- ifelse(dat1_ker$class == c & dat1_ker$Gender == gen &
      dat1_ker$enrolmt_3_35_yrs_status == 3, 1, 0)
    yind <- yind1 + yind2
    temp <- cbind(dat1_ker, xind, yind)
    R <- Rhat_function(temp) * 100
    se <- sqrt(varest(temp)) * 100
    rse <- se / R * 100.0
    res <- rbind(res, data.frame(class = c, gender = gen, r = R, se = se, rse = rse))
  }
}
print(res)

# =====
# END OF ANALYSIS
# =====
#
# Notes:
# - Sector: 1 = Rural, 2 = Urban
# - Gender: 1 = Male, 2 = Female
# - Enrollment Status Codes:
#   * 1 = Never enrolled
#   * 2 = Currently enrolled
#   * 3 = Enrolled in the past but not currently
# - Quintile Classes: 1 (Lowest) to 5 (Highest) based on UMPCE
# - Major_reason_not_enrolled: Code 15 excluded (completed education)
# - All estimates are in percentages
# - Standard errors calculated using stratified two-stage sampling design
# - RSE = Relative Standard Error = (SE / Estimate) × 100
#
# =====

```